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A Extended Related Work

Relating LLM outputs to pretraining data. Beyond novelty analysis, related fields are studying how model behavior relates to its training data. One such field is training data attribution (TDA). TDA studies how model behavior can be attributed to training samples through a causal lens by asking a counterfactual question: How would the model behavior change had the training samples not been part of the dataset (Hammoudeh and Lowd, 2024;

Deng et al., 2025)? If the change is large, the samples are highly influential. Works study whether models rely on relevant training samples for factual question-answering tasks (Akyürek et al., 2022; Chang et al., 2025), discover errors and biases in the model's learned patterns (Brunet et al., 2019; Wang et al., 2023), investigate how mislabeled data, outlier data, train-test domain mismatches, or simply which data samples influence learned model behavior (Koh and Liang, 2017; Pruthi et al., 2020; Park et al., 2023; Grosse et al., 2023; Choe et al., 2024). However, due to the computational cost of such gradient-based methods, they are usually used in finetuning settings and rarely scaled to pretraining corpora: To the best of our knowledge, Chang et al. (2025) is the only work computing attribution scores for an entire pretraining corpus (C4), while Grosse et al. (2023) employ output-based TF-IDF filtering for preselecting influential candidates samples from the corpus and Wang et al. (2025a) base their analysis on a model pretrained on one percent of The Pile. Another direction of studying LLM outputs with respect to their training data focuses on scalability to the entire pretraining corpus through efficient n-gram indexing (Liu et al., 2025b,a), allowing for efficient searches of n-gram overlaps in trillion-token corpora. Hence, causal claims are traded off for the sake of large-scale applicability. Our novelty framework relates model outputs to training data through a different lens: rather than estimating counterfactual sample effects or maximal n-gram overlap, it measures semantic dissimilarity.

B Filtered Dataset Sizes

For domain-specific novelty analysis (Section 4.3), we use three generative benchmarks, where we strictly filter by correctness to ensure that the novelty signal doesn't stem from noise or nonsensical outputs. Table 1 shows the sizes of the filtered datasets.

C Calibration via Human Baselines

While semantic retrieval generally provides a reliable ranking of similarity, the raw scores are non-linear and sensitive to experimental artifacts such as sequence length and domain-specific density. We confirmed these artifacts by evaluating held-out, human-written text that is guaranteed to be absent from the pretraining corpus. Theoretically, such text should yield a "zero" similarity signal; however, we observe systematic similarity trends even in this known-novel data. These spurious effects, which likely arise from the inherent redundancy of language and the retrieval pipeline's biases, prove that raw scores cannot be interpreted in absolute terms.

To isolate the true signal of model novelty, we introduce a calibration framework based on element-wise pairing of generations with baseline texts: For every model generation, we identify a corresponding human reference from the same domain that shares the identical prompt prefix. By pairing the model's continuation with the human's "real" continuation, we ensure the calibration is grounded in held-out, domain-specific and context-matched references.

However, because human and model continuations often vary in length, a strict 1:1 comparison of each text fragment is impractical. Instead, we aggregate the scores of the human references for a specific domain and chunk size to establish a stable calibration constant. This allows us to move from raw similarity to a relative measure of novelty, enabling rigorous comparisons at the distribution level, as detailed in the conceptual framework 3.

D Sufficiency of $n = 100$

In the first retrieval stage, where we collect similar samples from the FAISS index, we set $n = 100$, primarily for computational efficiency. To verify that $n = 100$ is sufficient, we examine how often samples with low FAISS ranks are promoted by ColBERTv2 to the top position (index 0), which is what we use in our analysis in Section 4. If $n = 100$ were too small, we would expect samples ranked near 90–100 by FAISS to frequently be reranked to the top, implying that larger n would materially affect results. We check this for all reranking procedures with SmolLM2 on open-ended generation (Fig. 2), using chunk size 500 to approximate whole-document reranking. The results (Fig. 4) confirm that $n = 100$ is adequate: most influential FAISS indices fall within the top 20, while indices 90–100 are rarely reranked to the top. Thus, larger n would have negligible impact on our findings. Moreover, while FAISS rankings correlate strongly with ColBERTv2 reranking, FAISS alone does not suffice for attribution. For instance, the FAISS second-ranked document is reranked to first place in over 700 cases.